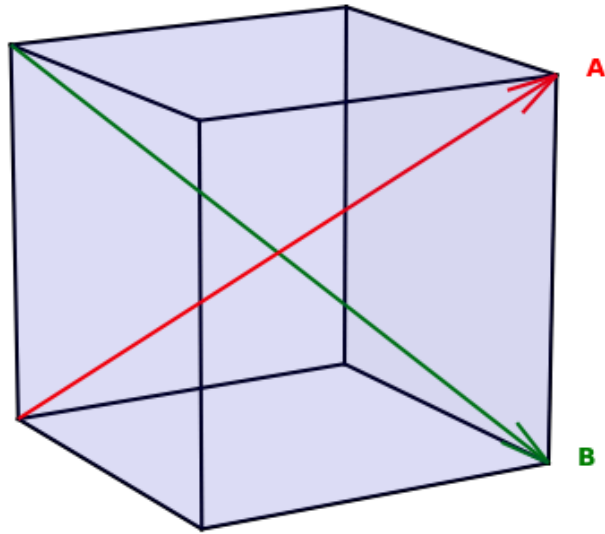


Find the angle between the body diagonals of a cube.

Solution:

The body diagonals $\mathbf{A} = [1, 1, 1]$ and $\mathbf{B} = [1, 1, -1]$



$$|\mathbf{A}| = |\mathbf{B}| = \sqrt{3}$$

$$\mathbf{A} \cdot \mathbf{B} = 1 + 1 - 1 = 1$$

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta = 3 \cos \theta = 1 \rightarrow \theta = \arccos\left(\frac{1}{3}\right) \approx 70.53^\circ \quad \square$$
